Independent Analytical Note

Derivation of the Moment of Inertia of a Solid Torus via the Cylindrical Shell Method

Md. Shahriar Parvez
Department of Mechanical Engineering
Bangladesh University of Engineering and Technology (BUET)
Dhaka, Bangladesh
sbijoy524@gmail.com

Abstract

Calculating the moment of inertia (I) of a solid torus about its central axis of rotational symmetry is traditionally performed using triple integrals in toroidal or spherical-polar coordinate systems requiring Jacobian determinants. In this note, I demonstrate an elegant alternative derivation by decomposing the torus into a continuum of concentric cylindrical shells. By reducing the geometry to a single-variable integration, we avoid multidimensional integration and arrive directly at the standard result:

$$I = M\left(R^2 + \frac{3}{4}r^2\right)$$

1 Geometric Setup & Definitions

Let us define a solid torus generated by revolving a circle of radius r around an axis lying in the same plane at a distance R ($R > r$) from the center of the circle.

- R: Major radius (distance from the center of the torus to the center of the tube)
- r: Minor radius (radius of the circular cross-section)
- M: Total mass of the uniform torus
- ρ: Mass density, where $\rho = \dfrac{M}{V_{\text{total}}}$

From Pappus's Centroid Theorem, the total volume of the solid torus is:

$$V = (2\pi R)(\pi r^2) = 2\pi^2 R r^2$$

2 Shell Decomposition & Differential Volume

Consider a circular cross-section of the tube parameterized by angle θ measured relative to the horizontal symmetry axis.
For an angular thickness $d\theta$, the cross-sectional slice yields two concentric cylindrical surfaces with radii:

- Outer radius: $r_2 = R + r\sin\theta$
- Inner radius: $r_1 = R - r\sin\theta$

The differential height element dh along the tube cross-section is:

$$dh = r\sin\theta\, d\theta$$

The differential volume dV contained between these two concentric cylindrical shells is given by the area difference multiplied by the vertical height differential:

$$dV = \pi(r_2^2 - r_1^2)\, dh$$

Substituting r_1, r_2, and dh:

$$r_2^2 - r_1^2 = (R + r\sin\theta)^2 - (R - r\sin\theta)^2 = 4Rr\sin\theta$$

Thus,

$$dV = \pi(4Rr\sin\theta) \cdot (r\sin\theta\, d\theta) = 4\pi Rr^2 \sin^2\theta\, d\theta$$

Converting $\sin^2\theta$ using the double-angle identity:

$$dV = 2\pi Rr^2(1 - \cos 2\theta)\, d\theta$$

Volume Verification Check: Integrating dV over $\theta \in [0, \pi]$:

$$V = \int_0^\pi 2\pi Rr^2(1 - \cos 2\theta)\, d\theta = 2\pi Rr^2 \left[\theta - \frac{\sin 2\theta}{2}\right]_0^\pi = 2\pi^2 Rr^2$$

This perfectly matches the known total volume of a torus.

3 Moment of Inertia Calculation

The moment of inertia dI for a thin cylindrical shell bounded by radii r_1 and r_2 with mass dM is:

$$dI = \frac{1}{2} dM(r_1^2 + r_2^2)$$

Where $dM = \rho dV = \dfrac{M}{V} dV = \dfrac{M}{2\pi^2 Rr^2} dV$.
First, evaluate $r_1^2 + r_2^2$:

$$r_1^2 + r_2^2 = (R - r\sin\theta)^2 + (R + r\sin\theta)^2 = 2(R^2 + r^2\sin^2\theta)$$

Now, substitute dM and $(r_1^2 + r_2^2)$ into the expression for dI:

$$dI = \frac{1}{2}\left[\frac{M}{2\pi^2 Rr^2} \cdot \left(4\pi Rr^2 \sin^2\theta\, d\theta\right)\right] \cdot 2(R^2 + r^2\sin^2\theta)$$

Simplifying the pre-factors:

$$dI = \frac{2M}{\pi} \sin^2\theta(R^2 + r^2\sin^2\theta)\, d\theta$$

$$dI = \frac{2M}{\pi} \left(R^2\sin^2\theta + r^2\sin^4\theta\right) d\theta$$

Integrating dI from $\theta = 0$ to $\theta = \pi$:

$$I = \frac{2MR^2}{\pi} \int_0^\pi \sin^2\theta\, d\theta + \frac{2Mr^2}{\pi} \int_0^\pi \sin^4\theta\, d\theta$$

Using standard definite integral properties:

1. $\int_0^\pi \sin^2\theta\, d\theta = \dfrac{\pi}{2}$
2. $\int_0^\pi \sin^4\theta\, d\theta = \dfrac{3\pi}{8}$ (evaluated via Beta/Gamma functions or reduction formulas)

Substituting these values:

$$I = \frac{2MR^2}{\pi}\left(\frac{\pi}{2}\right) + \frac{2Mr^2}{\pi}\left(\frac{3\pi}{8}\right)$$

$$I = MR^2 + \frac{3}{4}Mr^2$$

Factoring out M:

$$\mathbf{I = M\left(R^2 + \frac{3}{4}r^2\right)}$$

4 Conclusion

By breaking down the 3D solid torus into concentric cylindrical shells, this approach simplifies the math significantly. It bypasses the multi-variable coordinate transformations and Jacobian determinants that usually come with standard triple integrals, without losing sight of the underlying physics. Reducing the entire setup to a single-variable integration keeps the derivation clean, transparent, and easy to follow. Beyond just solving for the moment of inertia, this method serves as a practical example of how choosing the right geometric symmetry can make hard problems in classical mechanics much easier to manage.